Nila Karthikesan

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EDUCATION

University of Maryland, College Park — College Park, MD
B.S. in Computer Science, B.A. in Public Policy, and Minor in Engineering Technology — *Aug. 2020 - May 2024*
- **Director of Engineering, Society of Women Engineers:** Led technical workshops creating 50+ projects with JS, React, Node, REST APIs, and cloud-native backends.

Georgia Institute of Technology — Online
M.S. in Computer Science (Specialization: Artificial Intelligence) — *Aug. 2025 - Present*
- Contributed interactive Three.js visualization and data loading to GTSfM for hierarchical 3D reconstructions.
- Coauthored "Hierarchical Structure-from-Motion Scales FeedForward Reconstruction," presented at the ECCV 2026 SfM-DL workshop.

EXPERIENCE

Software Engineer, Platform Infrastructure — Oct. 2025 - Present
Pinterest — *New York, NY*
- Building internal cloud governance and automation for the **Security Platform (PlatSec)** org, managing and securing **thousands of GCP projects and AWS resources** via **Terraform IaC**.
- Designed an **S3-backed Terraform remote state** and workspace strategy with label-to-tfvars and declarative import automation, enabling safe migration of existing projects with **no-op plans** and no destructive drift.
- Implementing centralized org audit logging (GCP log sinks -> **Pub/Sub -> Cloud Function -> S3 -> Gravwell**) for low-latency, scalable security analytics across distributed systems.

Software Engineer — May 2024 - Oct. 2025
Geico — *New York, NY*
- Led development of a **full-stack internal platform** managing GEICO's server inventory, exposing **100,000+ server records** via a **React UI** and **GraphQL API** to **500+ engineers**.
- Reduced server query by **40%** through improved data modeling, caching, API design across PostgreSQL, GraphQL resolvers.
- Migrated reporting workflows from **Power BI to PostgreSQL/Python**, cutting data processing time by **50%** and improving reliability of infra telemetry pipelines.

Software Engineering Fellow — Dec. 2023 - Jan. 2024
Palantir Technologies — *New York, NY*
- Developed a backend for a proxy service handling **10,000+ IPs** & **100+ monthly users** on an AI platform, focusing on reliable APIs and observability in production.
- Improved project load time by **80%** (minutes -> seconds) by containerizing services with **Docker & Kubernetes** and optimizing I/O paths.

Software Engineering Intern — May 2023 - Aug. 2023
Comcast — *Washington, DC*
- Built a **full-stack content moderation platform** using **React.js, Flask, and OpenAI's CLIP**, integrating ML models in production to classify and flag unsafe media across creator content.
- Implemented containerized microservices with **Docker & Kubernetes on AWS**, improving horizontal scalability by **40%** and supporting concurrent moderation workloads.
- Integrated **Elasticsearch** to power low-latency search and review of flagged items, achieving an **8x reduction in retrieval time** and boosting moderator efficiency.

PROJECTS

Grasp Recovery under Delayed Feedback — Oct. 2026 - Present
MuJoCo simulation pilot | *Repository*
- Built a standalone MuJoCo grasp-transport-place task with bit-exact replay; collected 400 pilot episodes and 88,728 matched recovery branches under hidden mass/friction and independently varied observation age and response delay.
- Evaluated recovery selectors on 94 held-out episodes with episode-clustered bootstrap intervals; latency-aware selection showed no completion advantage over its latency-agnostic baseline. Preliminary, simulation-only results.

CLIP-Based Content Moderation System | *Python, FastAPI, Supabase, OpenAI CLIP, Docker, React* — 2024
- Developed a full-stack ML system analyzing video/image embeddings with **OpenAI's CLIP** to detect unsafe content, using **FastAPI, Supabase/Postgres, and Dockerized microservices** for scalable moderation pipelines.
- Built cosine-similarity-based embedding search and React dashboard for reviewers, enabling real-time classification, flagging, and explainable model-inference UX.

Search Engine Optimization Evaluation Agent | *Python, LangChain, OpenAI, Supabase, Streamlit* — 2025
- Created an **LLM-based evaluation agent** using LangChain to generate and rank responses from **ChatGPT/Claude/Perplexity**, storing structured metrics in **Supabase** for analysis and iteration.
- Designed a **Streamlit dashboard** to visualize rankings and quality scores, showcasing agentic orchestration similar to search/recommendation model tuning workflows.

TECHNICAL SKILLS

Languages: Python, Java, Go, C/C++, SQL (Postgres), JavaScript, Rust, R, HTML/CSS
Frameworks: React, Node.js, Flask, FastAPI, GraphQL, Temporal, JUnit
Cloud & Infra: AWS (EC2, S3, RDS, Lambda), GCP (Logging, Pub/Sub, Cloud Functions), Terraform, Docker, Kubernetes, Git
Libraries: Pandas, NumPy, Matplotlib, SciPy